

HybridViT-CAB: a vision transformer and convolutional attention network for precision weed detection in agricultural systems

Received: 4 January 2026

Accepted: 9 April 2026

Published online: 21 April 2026

Cite this article as: Yenna G.R., Kumari N.V., Chouhan A.S. *et al.* HybridViT-CAB: a vision transformer and convolutional attention network for precision weed detection in agricultural systems. *Sci Rep* (2026). <https://doi.org/10.1038/s41598-026-48652-7>

Geetha Reddy Yenna, Nalli Vinaya Kumari, Arun Singh Chouhan, Syed Umar & Sandhya Rani Macha

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

HybridViT-CAB: A Vision Transformer and Convolutional Attention Network for Precision Weed Detection in Agricultural Systems

***Geetha Reddy Yenna,**

Associate Professor, Dept of Computer Science Engineering, Malla Reddy Engineering College for Women, Maisammaguda, Dhulapally, Post via Kompally, Secunderabad-500100, Telangana, INDIA.
Email: geethareddy1204@gmail.com

Nalli Vinaya Kumari,

Associate Professor, Dept of CSE - Artificial Intelligence and Machine Learning, Malla Reddy Engineering College for Women, Maisammaguda, Dhulapally, Post via Kompally, Secunderabad-500100, Telangana, INDIA
Email: v.vinayakumari@gmail.com

Arun Singh Chouhan,

Associate Professor, CSE, Malla Reddy University, Maisammaguda, Dhulapally, Hyderabad, Telangana 500100, India.
arunsingh@mallareddyuniversity.ac.in

Syed Umar,

Professor, Dept. of CSE, Malla Reddy (MR) Deemed to be University, Maisammaguda, Medchal-Malkajgiri, Telangana-500100, India. syedumar@mrec.ac.in

Sandhya Rani Macha,

Professor, Department of Pathology, Malla Reddy Medical College for Women, Malla Reddy Vishwavidyapeeth, Suraram, Hyderabad 500055, Telangana, India.
Email: sandhyarani.macha@mrvv.edu.in

Abstract.

In this study, we propose a new hybrid deep learning architecture combining Vision Transformers (ViT) with Convolutional Attention Blocks (CAB), specifically designed for automated weed detection in precision agriculture. We introduce a novel coupled approach that combines transformer-based global contextual modelling with channel-wise attention-based local spatial feature extraction to classify images into four classes: broadleaf weeds, grass, soil, and soybean crops. Following training on a balanced dataset of 4,795 samples resized to 224×224 pixels, the proposed model achieves an overall AUC of 0.97. While the ViT-only configuration achieves the highest classification accuracy (91%), the hybrid ViT-CAB architecture offers superior interpretability via Grad-CAM visualisations and more balanced per-class performance across morphologically similar vegetative categories, making it the preferred choice when explainability and class-balanced prediction are priorities for precision agricultural deployment. This compact model (412K parameters, 1.57 MB) processes each image in 141ms per step and is thus practical for real-time use in intelligent weed management systems aiming at sustainable farming mechanisms.

Keywords. Weed detection, ViT, Attention network, deep learning, ablation study.

1. Introduction

Farmers around the world are facing greater challenges than ever to provide food security for a population predicted to reach 9.7 billion by 2050 [FAO, 2023], while also responding to demands for environmental sustainability and to the effects of climate change. Weed infestation is one of the most significant biotic constraints on global crop production, with yield losses of 34% to 40% estimated in major agricultural systems Murad et al. (2023). Broadcast application of synthetic herbicides is the primary approach to weed management worldwide. While effective, it may lead to a range of serious environmental problems, including soil degradation, pollution of water bodies, herbicide-resistant weed biotypes Murad et al., (2023), and impacts on non-target organisms. Growing regulatory restrictions on the non-target use of chemical herbicides and increasing market demand for organic and sustainably produced agricultural products have driven the need for the development of economically viable yet ecologically sound alternative weed control methodologies.

The paradigm of precision agriculture thus evolved, reliance on advanced technologies, remote sensing, IoT sensor-based applications, robotics, and artificial intelligence made it possible to optimise on-site inputs while reducing carbon footprints and maintaining production. Along these lines, site-specific weed management (SSWM) is one of the most impactful application areas for applying herbicides or performing mechanical weed control solely in infested areas, leading to a substantial reduction (up to 70–90%) in chemical use through SSWM compared to conventional broadcast spraying methods Pai et al. (2024). Inevitably, the reliable performance of SSWM systems is contingent on the ability to detect and classify, in real time, crop plants and broad-spectrum weed species, as well as bare soil, across a wide range of field environments—including different conditions of illumination, onto genic stages of growth, occlusion and background (Pai, D. G., et al, 2024). The significant morphological overlap between many weed types and crop plants, and the high levels of within-species phenotypic plasticity in response to environmental conditions, pose significant hurdles for automated classification systems.

One of the most frequently used methods for the detection of weeds at an early stage was based on conventional machine learning techniques such as Support Vector Machines (SVM) Murad, N. Y., et al (2025), Badhan, S., et al (2021), Random Forests, and k-Nearest Neighbors (k-NN) Etienne, A., and Saraswat, D. (2019, May). These methods were found to be effective for the detection task under controlled conditions. However, they were found to lack generalization ability for the task in real-world conditions due to various factors such as background clutters and varying illumination conditions. In feature engineering, human expertise may play an important role in engineering discriminative features for the task at hand. However, feature engineering may not capture hierarchical patterns at various scales that may play an important role in discrimination between crops and weeds. Additionally, the conventional methods were found to face challenges due to the variability of plant phenotypes at various stages of growth, seasonal variations, and geographical variations.

Deep learning methods, especially Convolutional Neural Networks (CNNs), dos Santos Ferreira, A., et al (2017), Al-Badri, A. H., et al (2022), Jin, X., et al (2022), and Jin, H., et al (2025) have transformed many fields of computer vision application domains ranging from medical imaging, self-driving cars, face recognition, and plant pest diseases automation and management. Traditional methods largely depend on manual feature engineering, while CNN-based solutions achieve significant improvements by automatically learning hierarchical feature representations from pixel values. Moreover, traditional architectures such as AlexNet, VGGNet, ResNet, and

EfficientNet have also been applied to weed classification. Accuracies greater than 90% have been reported on weed benchmark datasets using these networks (Goyal, R. et al., (2025); Tummapudi, S. et al., (2023)). These models leverage local spatial information via learned convolutional operations with ever-increasing receptive fields, which facilitate learning helpful cues such as texture, edges, and shape to separate plant species. However, conventional CNNs are inherently limited by their local receptive fields to modelling long-range dependencies and global contextual interactions. Consequently, they are unable to adequately represent holistic plant architecture and plantation arrangement, which are essential for crop–weed separation under rich-field conditions. Many of the researchers have used YOLO based models Kanade, A. K., et al (2025), Lin, J., et al (2025), Khan, A. T., et al (2025) and Deng, B., & Lu, Y. (2025) and achieved better results.

An essential component of the recent paradigm shift in computer vision, spurred by ViT, has been the adaptation of self-attention mechanisms for image-based tasks. In contrast to CNNs that work their way up through many stages of hierarchical convolutions with local receptive fields, Vision Transformers divide images into patches, allowing multi-head self-attention mechanisms to model relationships between all patches at once, leading to accurate global context modelling from the first layers of processing. This architecture has achieved state-of-the-art performance on large-scale image classification across ImageNet and various benchmark datasets. Since Vision Transformers utilise the self-attention mechanism, which has high model capacity for long-range dependencies, they may be suitable for agricultural applications where distinguishing between morphologically similar species based on local textures often necessitates combining global structural patterns, e.g., leaf arrangement (phyllotaxis), whole-plant architecture, and growth habits.

Although the performance of ViT in capturing the global context is excellent Wang, A., et al (2019) it tends to ignore local features, thus, missing detailed texture with fine details that are important to distinguish plant species such as the leaf surface characteristics, venation patterns, or even cellular level features (depending on the appropriate resolution of the imaging system) on the same axis (such as texture) where the plant structures may be indistinguishable. On the other hand, CNNs have great potential to reveal local discriminative features through embedded convolutional operations. However, they might not necessarily use these features to build a holistic understanding of plant morphologies and the environment. Recent studies by Rana et al. (2026) have shown that attention-based and geometry-aware architecture can be effectively used to overcome detection failures caused by occlusion and orientation in agricultural images. This further emphasizes the importance of using attention-based architecture in solving the weed detection problem. The advantages of global and local feature extraction are another motivation towards developing a hybrid architecture that can take advantage of both paradigms.

We want to lead attention mechanisms for augmenting convolutional architectures, enabling the model to focus on the essential features and ignore irrelevant background information. The channel attentions, such as those used in Squeeze-and-Excitation (SE) blocks, can play a significant role in recalibrating feature maps by learning what to attend to in feature space. This is complemented by spatial attention mechanisms that learn "where" to attend in spatial dimensions. By combining attention mechanisms with convolutional layers to implement convolutional attention blocks (CAB), our approach achieves greater discriminative power while maintaining low computational cost, a crucial property for many agricultural applications where target vegetation must be distinguished from complex backgrounds.

Contributions:

- We propose HybridViT-CAB, compact hybrid architecture (412K parameters, 1.57MB) specifically designed for small agricultural datasets where standard data-hungry ViTs fail to generalize, addressing the practical constraint of limited labeled field data in precision agriculture.
- We introduce a convolutional patch embedding strategy replacing standard linear projection, providing built-in translation equi variance that is particularly beneficial for weed images with variable orientation and scale.
- We conduct a systematic six-variant ablation study that quantifies the individual contribution of transformer depth, channel attention, and architectural fusion for weed classification, providing design guidelines for future agricultural deep learning systems.
- We demonstrate that a model of only 412K parameters achieves AUC of 0.97 with 141ms inference time, confirming suitability for real-time edge deployment in resource-constrained agricultural environments.
- We provide Grad-CAM explainability analysis confirming that the model attends to botanically meaningful plant structures rather than background artifacts, a critical requirement for farmer-trusted AI systems."

2. Related work

2.1 Traditional Machine Learning Approaches

Traditional machine learning algorithms have dominated the early research in automated weed detection. Islam et al. (2021) [2] proposed SVM, random forest, and KNN algorithms for automated weed detection in a chili crop using images obtained from a UAV. They achieved a maximum accuracy of 0.96 for the random forest algorithm. However, these algorithms do not have the potential to incorporate spatial and hierarchical features directly from images. Urmashev et al. (2021) [5] proposed a hybrid approach that incorporates both machine and deep learning algorithms after obtaining 1,000 samples. They achieved a classification accuracy of 0.83. Badhan et al. (2021) [6] proposed a real-time approach for automated weed detection in a crop field using a combination of machine learning and stereo vision. Etienne and Saraswat (2019) [15] proposed a machine learning approach for automated weed detection in a crop field. They incorporated a variety of sensors on a UAV for automated detection. They demonstrated the potential of automated detection in a crop field. However, classical descriptors do not have the potential for automated detection in a crop field. In another study, Dadashzadeh et al. (2020) [36] implemented an ANN model and optimized video samples using a particular optimizer. They achieved an accuracy of 0.88 for the classification of weeds in rice fields. These conventional methods have shown promise in a controlled environment but lack sufficient generality in the presence of a cluttered background, varying illumination conditions, and varying shapes of weeds.

2.2 CNN-Based Deep Learning Models

The introduction of CNNs marked a significant advancement in weed detection by enabling automatic hierarchical feature learning from raw pixel values, eliminating the need for manual feature engineering. Yu et al. (2019) [9] trained a CNN model on binary images for the identification of perennial Ryegrass using 15,486 negative and 17,600 positive images, achieving an accuracy of 0.99. In a companion study, Yu et al. (2019) [10] also utilized a CNN to identify broadleaf weeds in turf grass, likewise achieving 0.99 accuracy. Alam et al. (2020) [11] fine-tuned a computer vision approach to identify weeds in real time, training the model on a newly created dataset and validating it in a live field environment. Razfar et al. (2022) [13] developed a deep learning model for weed detection specifically in soybean crops using 400 samples with

1,536 segmentations to train a five-layer CNN, achieving an accuracy of 0.97. This work is particularly relevant to the present study as it uses the same crop domain. Mahmudul Hasan et al. (2022) [24] trained a CNN on a dataset of 5,660 images after augmentation, using 90% for training and achieving an accuracy of 0.98 with batch sizes of 16 and 32. Fan et al. (2023) [31] implemented a deep learning model combining attention mechanisms with bilinear interpolation, training on 4,694 samples and achieving an accuracy of 0.98 for weed detection in cotton fields at the seedling stage. Osorio et al. (2020) [8] used a combination of deep learning and machine learning including SVM and YOLO models trained on multispectral images, achieving accuracy greater than 80% for weed detection in lettuce crops. While CNNs demonstrated strong performance on local spatial features, their inherent limitation of fixed local receptive fields prevents them from modeling long-range dependencies and global structural patterns essential for distinguishing morphologically similar plant species.

2.3 Detection-Based Models (YOLO and R-CNN)

Object detection frameworks including YOLO and R-CNN variants have been widely adopted for weed detection due to their ability to simultaneously localize and classify multiple plant instances within a single image. Wang et al. (2022) [14] created a new dataset comprising 14,035 images from 25 weed samples and trained YOLO and R-CNN models on it, achieving 91% accuracy. Jin et al. (2022) [28] used YOLO-v3, CenterNet, and R-CNN with a confidence threshold, achieving 0.97 accuracy on test data with YOLO-v3 performing best among the evaluated models. Mwitwa et al. (2024) [30] used the YOLO-4 model where video data was first converted into image samples at 14 frames per second, generating 13,000 weed class RGB samples and 67,000 augmented samples for training, achieving an accuracy of 0.93 and a mean average precision of 0.88. Sapkota et al. (2022) [22] used synthetic data and UAV-based images for weed and biomass detection in cotton crops, augmenting samples with a GAN before training an R-CNN model for classification of grass and weeds with bounding-box detection. Veeranampalayam Sivakumar et al. (2020) [23] worked on five models including VGG, ResNet, and Inception with a focus on imbalanced data, finding that ResNet achieved better accuracy among the evaluated architectures. Li et al. (2024) [27] trained a deep learning model on 9,278 manually annotated RGB samples using R-CNN, YOLO-R-CNN, and a Transformer model with spatial attention, increasing accuracy from 0.85 to 0.88 for winter wheat weed detection. Islam et al. (2020) [26] used UAV images captured in RGB format and trained a machine learning model, achieving an accuracy of 0.98 for weed detection in chilli fields. Rahman et al. (2023) [35] used 13 deep learning models trained using UNET, VGG, and YOLO architectures on 848 samples of a three-class weed dataset, achieving an average accuracy of 0.78. Asad and Bais (2020) [34] used a two-step approach where background and foreground samples were first segmented using maximum-likelihood encoder-decoder classification and then manually labeled weed samples were used to train UNET and VGG models.

2.4 Attention-Based and Hybrid Transformer Architectures

Recent developments have included the application of attention mechanisms and Vision Transformers in weed detection systems, based on their capacity for modeling long-range dependencies and global contextual relationships, which cannot be achieved by CNNs. Zhang et al. (2023) [19] used VGG models as a backbone and global average pooling for feature extraction in weed detection in soybean seedling stages, achieving an accuracy of 0.99. Selvi et al. (2021) [20], Junior et al. (2021) [16], and Punithavathi et al. (2023) [17] used deep learning-based weed detection systems in which optimizing farmland fertility helps in improving key parameters of their models. Bah et al. (2018) [12] proposed an unsupervised learning-based weed

detection system using UAV imagery and CNN classification, marking one of the earliest attempts in this direction for minimizing annotation requirements.

Garibaldi-Márquez et al. (2025) [38] presented a transformer architecture based on the SegFormer approach for site-specific weed detection, achieving 0.94 accuracy. Modi et al. (2023) [25] presented an automated framework for identifying weeds in sugarcane crops using the deep learning approach. The framework is optimized for practical applications. Wang et al. (2022) [43] presented an experimental study of various YOLO-based approaches like YOLOv3, YOLOv5s, YOLOv7, YOLOv8s, and YOLOv9s for detecting weeds in cotton field crops. The accuracy of the YOLOv8s approach is 0.93. Allmendinger et al. (2025) [45] assessed the capability of YOLO-based and transformer-based object detectors for real-time weed detection in field conditions, with YOLOv9c achieving 0.84 accuracy. Rana et al. (2026) [47] demonstrated that attention-based and geometry-aware architectures can effectively mitigate occlusion and orientation-induced detection failures in agricultural imagery, further motivating the integration of attention mechanisms into weed detection frameworks.

Despite these advances, no prior work has systematically evaluated compact hybrid ViT-CAB architecture specifically on a small balanced four-class soybean weed dataset with explicit six-variant ablation validation of each architectural component. Existing hybrid models use large datasets, focus on detection rather than classification, or do not provide systematic component-wise ablation evidence. This motivates the present study which addresses these gaps through a deployment-ready compact model validated on a controlled balanced dataset with full architectural transparency.

3. Methodology

This paper presents a novel hybrid deep learning architecture for the proposed weed detection system, integrating ViT components with CAB to accurately and reliably classify multiple weed classes as shown in Figure 1. This approach combines the long-range contextual awareness of transformer-based attention models with the local feature-extraction power of CNNs to develop a unified framework for precision agriculture.

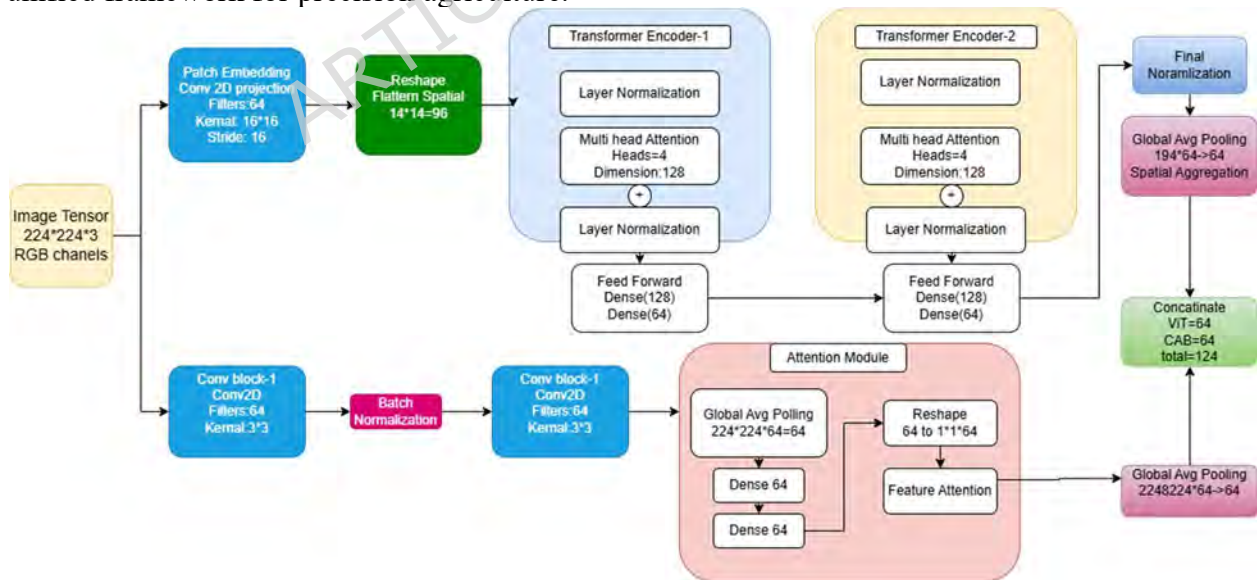


Figure 1: Complete architecture of the proposed model

Dataset Preparation and Preprocessing

The dataset used in this study is the publicly available Weed Detection in Soybean Crops dataset obtained from Kaggle (<https://www.kaggle.com/datasets/fpeccia/weed-detection-in-soybean->

[crops](#)). The original dataset contains 21,035 field images in TIFF format across four classes: broadleaf weeds, grass, soil, and soybean crops. Images were captured using an RGB camera mounted on agricultural equipment at ground level under natural field illumination conditions in soybean production fields in Brazil. To create a class-balanced dataset and prevent dominant classes from biasing model training, a maximum of 1,200 samples per class was selected using a systematic traversal of the class folders. This resulted in a final dataset of 4,795 images distributed as: Broadleaf (1,195), Grass (1,200), Soil (1,200), and Soybean (1,200). The Broadleaf class contains 1195 images as the original dataset contained fewer than 1,200 TIFF files for this class. All images were converted from TIFF to JPEG format using the Python Imaging Library (PIL) and resized to 224×224 pixels using bilinear interpolation during the conversion pipeline. Pixel intensity values were normalized to the range [0,1] by dividing each RGB channel by 255. The dataset was divided into training (80%) and held-out test (20%) subsets using stratified sampling with random seed 42 for reproducibility, ensuring equal class proportions in both subsets. This resulted in 3,836 training images and 959 test images as shown in Table 1.

Table 1. Class-wise distribution of the balanced dataset after stratified splitting.

Class	Training (80%)	Testing (20%)	Total
Broadleaf	956	239	1,195
Grass	960	240	1,200
Soil	960	240	1,200
Soybean	960	240	1,200
Total	3,836	959	4,795

Hybrid Architecture Design

The hybrid architecture is composed of three basic modules: a patch embedding module, transformer encoder blocks, and convolutional attention blocks, all running in parallel (fused via feature fusion as in (1)). Unlike ViT implementations that use linear embedding as the patch embedding module, this architecture uses a convolutional projection layer. Remember specifically that, through a 2D convolutional layer with 64 filters, a kernel size of 16×16, and a stride of 16, we obtain non-overlapping patches from the input image and project them into a 64-dimensional space. This convolutional approach provides built-in translation equivariance with fewer learnable parameters than the fully connected patch embedding layers. Then, the feature maps are reshaped into a sequence for conversion into patch embeddings for transformer processing.

For the transformer encoder pathway, two stacked encoder blocks are used, where each implements the standard transformer architecture (i.e., multi-head self-attention mechanisms). The input sequence is first-layer normalised before performing multihead attention with four heads and a 128-dimensional key dimension. Using multihead attention, the model can learn diverse relational patterns across various spatial locations in the input image, allowing it to capture long-range dependencies that are particularly helpful for differentiating subtle morphological differences among weed species. The attention result is residually connected to its original input and then goes through another layer of normalisation. The normalised features are passed through a feed-forward network consisting of two dense layers — the first layer expands to 128 units with ReLU activation, and the second projects back down to the original dimension of the embedding. A second residual connection combines (adds) the output of the feed-forward mechanism with its input so gradients can flow through the stack unchanged and enable us to

train deep architectures. The global contextual feature vector produced by the ViT pathway is formally defined in Equation 1, where layer normalisation, multi-head self-attention with four heads, and position-wise feed-forward operations are applied sequentially with residual connections across two stacked transformer encoder blocks..

$$f_{vit} = \frac{1}{N} \sum_{i=1}^N LayerNorm \left(Z_{i-1} + FFN \left(LayerNorm \left(Z_{i-1} + softmax \left(\frac{(zw^q)(zw^q)}{\sqrt{d_k}} \right) (zw^v) \right) \right) \right) \quad (1)$$

Convolutional Attention Block works directly on the original input images and pays the price for local features extraction capabilities in a complementary way with the help of equation (2). This branch starts with two consecutive 3x3 convolutional layer operations. This produces 64 features with ReLU and batch normalization. The convolutional operations capture spatial patterns, edge features, and texture features, which are useful features for distinguishing weeds. The attention mechanism of this block is the squeeze-excitation approach. In the squeeze-excitation approach, the spatial dimensions are squeezed using global average pooling. Then two fully connected layer operations follow. The first layer has 64 units with ReLU activation for non-linear transformation. The second layer has 64 units with sigmoid activation for channel-wise attention weights. The spatial dimensions are regained by reshaping the weights and multiplying them element-wise with the convolutional features. This allows the model to attend to the discriminative features while ignoring the irrelevant background features. Global average pooling is applied to the features to get a compact feature.

$$f_{cab} = \frac{1}{HW} \sum_i \sum_j [C_3[i, j, :] * \sigma(w_2 \cdot ReLU(w_1 \cdot GAP(C_3) + b_1) + b_2)] \quad (2)$$

Feature Fusion and Classification

The ViT path provides a set of global context features, and the Convolutional Attention Block provides local spatial features of the same image (wide seeing and local eyes). These two-dimensional features are fused and concatenated to form a joint representation. A fully connected layer of size 128 is applied to the concatenated feature vector, enabling the network to have extra non-linear transformation capacity to learn the complex decision boundaries described in equation (3). It is using a dropout layer with a 0.4 rate to avoid overfitting. Randomly switching off neurons during training will improve generalisation to unseen test samples. For the last classification layer, a dense layer with softmax activation is used, yielding a probability distribution over all weed classes, with the number of output units equal to the number of weed classes in the dataset.

$$\hat{y} = softmax(W_{out} \cdot Dropout(ReLU(w_{dense} \cdot [f_{vit} \oplus f_{cab}] + b_{dense}), 0.4) + b_{out}) \quad (3)$$

Training Configuration and Optimization

The model is compiled using the Adam optimiser — an algorithm that computes individual adaptive learning rates for different parameters from estimates of the first and second moments of the gradients. Sparse categorical cross-entropy is an appropriate loss function for multi-class classification when labels are integers. The training is carried out for 25 epochs, using 32

samples to balance memory constraints and convergence speed. We use a data pipeline with prefetching and shuffling (buffer_size=1024) to randomise data and maximise GPU usage. During training, classification accuracy is used as the primary metric to evaluate model performance on the training and validation datasets, thereby enabling assessment of learning progress and potential overfitting.

Training was conducted for a maximum of 25 epochs with early stopping applied with a patience of 5 epochs based on validation loss to prevent overfitting. A cosine annealing learning rate schedule was applied, decaying the initial learning rate of 1×10^{-3} to a minimum of 1×10^{-5} over the training duration to improve convergence stability

We acknowledge that the individual components of Hybrid ViT-CAB, namely squeeze-excitation style channel attention and feature concatenation fusion, are established operations in the deep learning literature. The novelty of this work lies not in proposing new mathematical formulations but in the specific architectural coupling decision, the systematic ablation validation, and the demonstration of a practical deployment-ready solution for small-scale agricultural datasets where existing hybrid architectures have not been explicitly evaluated.

All experiments were conducted on Kaggle's cloud computing environment using an NVIDIA Tesla T4 GPU with 16GB VRAM. The model was implemented using Tensor Flow 2.x with Python 3.9. The reported inference time of 141ms per step corresponds to this hardware configuration with a batch size of 32.

4. Results and analysis

The hybrid ViT-CAB model is highly learned and also light (1.57 MB), requiring merely 412,676 trainable parameters after 25 epochs of training. The model was trained and evaluated on the complete balanced dataset of 4,795 images with 3,836 training samples and 959 held-out test samples. The model converges well in its initial epochs, with the first epoch reaching a training accuracy of 55.09%, increasing to 74.21% for the second epoch, indicating that the hybrid architecture propagates appropriate gradients across its layers. Two evaluation sets are referenced throughout this section. The internal validation set refers to the 20% subset of training data used to monitor training progress and apply early stopping during model training. The held-out test set (n=959 images, 20% of total dataset) was completely withheld during training and is used for all final performance reporting. Accuracy values reported during training epochs (including the peak of 90.20% at epoch 23) refer to the internal validation set. All metrics in Table 2 including the 88% overall accuracy refer to the held-out test set.

Progressive weed classification capabilities are shown in the refining learning curve, with a peak validation accuracy of 90.20% at epoch 23, and a corresponding training accuracy of about 90.08%. This close training and validation performance, along with the small dropout regularisation (0.4), suggests an appropriate model complexity without significant overfitting. The loss metrics support this view, with the training loss dropping from 0.9820 to 0.2301, and the validation loss stabilising at around 0.23-0.31 in subsequent epochs.

Yet across epochs 11–22, the accuracy on the validation set varied (83.21%–89.47%), indicating the model was likely struggling to learn complex decision boundaries where some morphologically similar weed species occur. The computational performance for the training is illustrated, with the first epoch running for 63 seconds and the remainder stabilising at about 17 seconds/epoch (140-141ms/step). This reflects the fast ORB pipeline and the hybrid architecture designed for practical agricultural deployment.

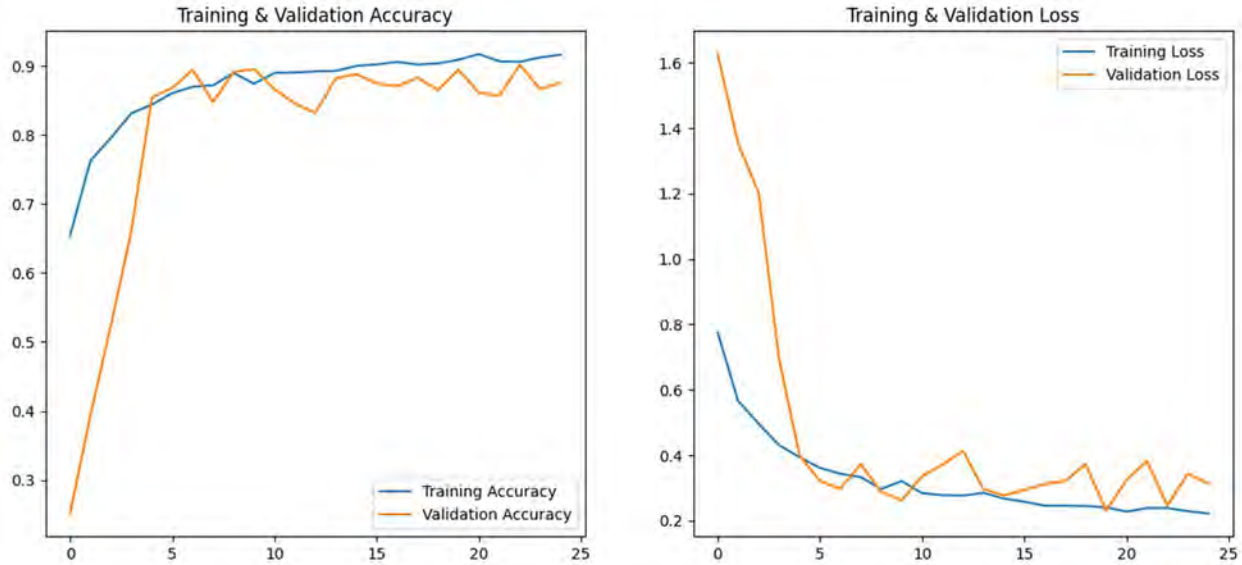


Figure 2 learning curves of the proposed model

This is followed by a hybrid ViT-CAB model that classifies weeds into 4 classes, achieving an overall classification accuracy of 88% on the held-out test set of 959 samples. Figure 3 — Confusion matrix for such a distinct classification, the model achieves perfect classification of soil samples (Precision, Recall and F1-score = 100% (239/240)) from table 2. This indicates that the model excels at distinguishing non-vegetation cover from vegetation through the interaction between global context features and local spatial information.

Table 2 consolidated metrics

Class	Test Samples	Precision	Recall	F1-Score	AUC
Broadleaf	239	0.81	0.83	0.82	0.97
Grass	240	0.94	0.68	0.79	0.97
Soil	240	1	1	1	1
Soybean	240	0.79	1	0.88	1
Macro Average	959	0.89	0.88	0.87	0.97
Overall Accuracy					88%

About soybeans, we predicted all soybeans (240), and so our recall here is also 100%. But our precision is 79% (F1-score: 0.88), which means only a few false-positive predictions, which are only for broadleaf weed (31 misclassifications), were made by our model. The grass category has the most incredible precision (94%), which is encouraging and builds confidence in the prediction (false positives are low). With a recall of 68% (F1-score: 0.79), the algorithm fails to identify all actual grass instances, misclassifying 46 samples as broadleaf and 31 as soybean. Balanced broadleaf weed detection performance is 81% precision and 83% recall (F1-score: 0.82), attaining 199 correct out of 239.

With precision = 0.89, recall = 0.88, and F1-score = 0.87 for all classes, it was found that the predictive performance for all classes was almost at par, which signifies the utility of the model for precision farming. Also, the extent of confusion between classes is low for morphologically

distinct classes (soil and vegetation), and the maximum extent of confusion occurs between morphologically similar classes at the highest level of vegetation (broadleaf, grass, and soybean).

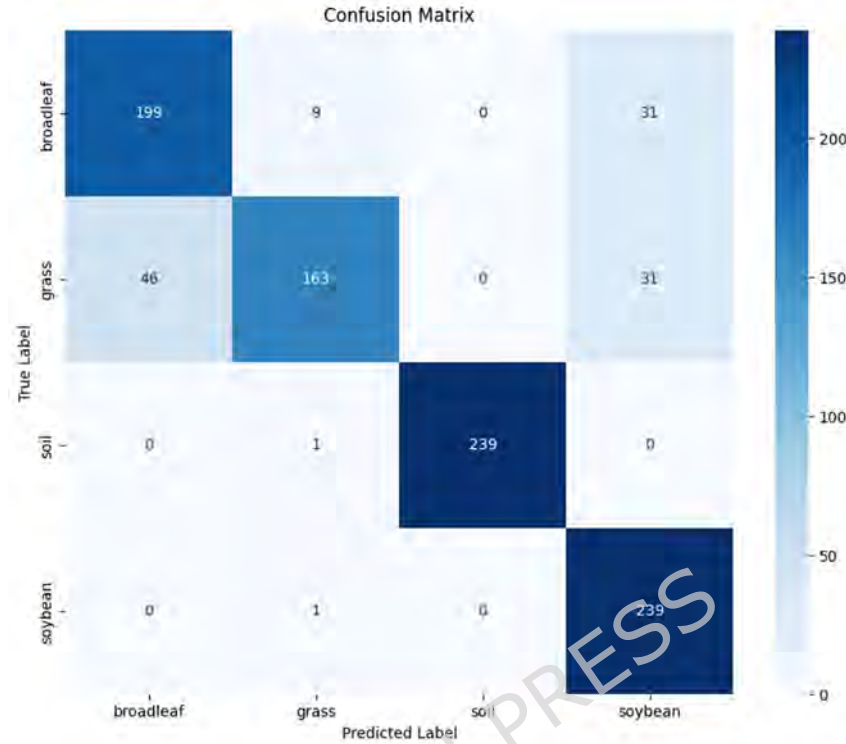


Figure 3: confusion matrix proposed model

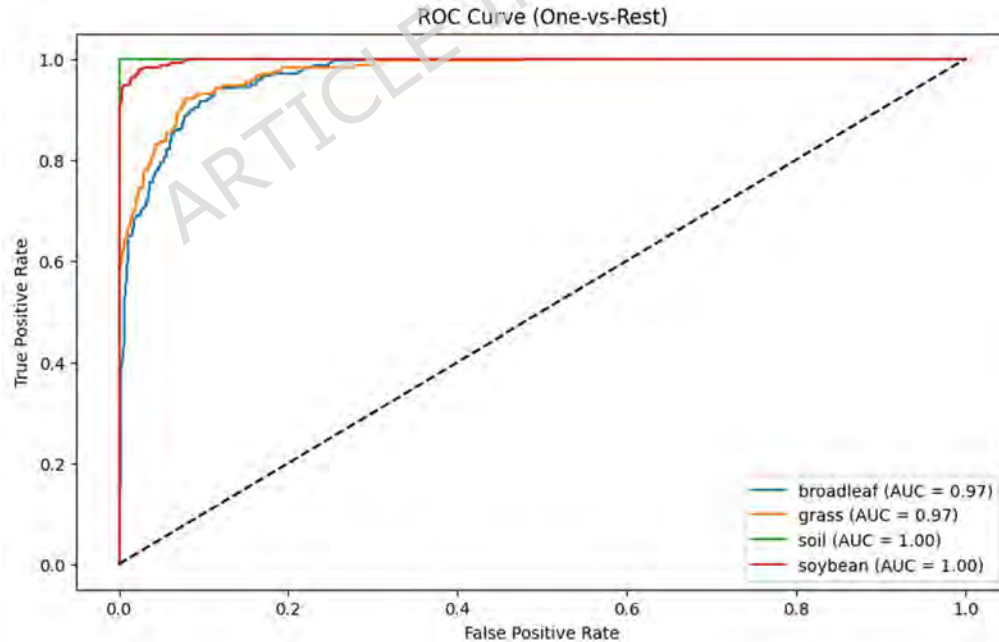


Figure 4: ROC curve of the proposed model

The Receiver Operating Characteristic (ROC) curves show the discriminative ability of the model for all weed categories using the one-vs-rest classification method. The AUC metric in Figure 4 is 1.00 for the soil and soybean categories, suggesting an extraordinarily high

classification performance and perfect separation from the other classes across all decision thresholds. Both broadleaf and grass categories achieve AUCs of 0.97, suggesting reliable discrimination, although elements of these categories resemble crop vegetation.

ROC curves for all classes show characteristic steep rises in the upper-left quadrant, indicating highly accurate favourable rates at very low falsefavourable rates. This behaviour affirms that we can still achieve good classification boundaries with larger variations in the confidence threshold using our model. AUC of 1 about the soil class indicates the model's ability to classify non-vegetative surfaces based on different textures and spectral characteristics. The high AUC values achieved across all categories, especially in the 1-M cultivars, confirm that the hybrid ViT-CAB architecture can learn both global contextual patterns and local discriminative features necessary for precision weed management applications.

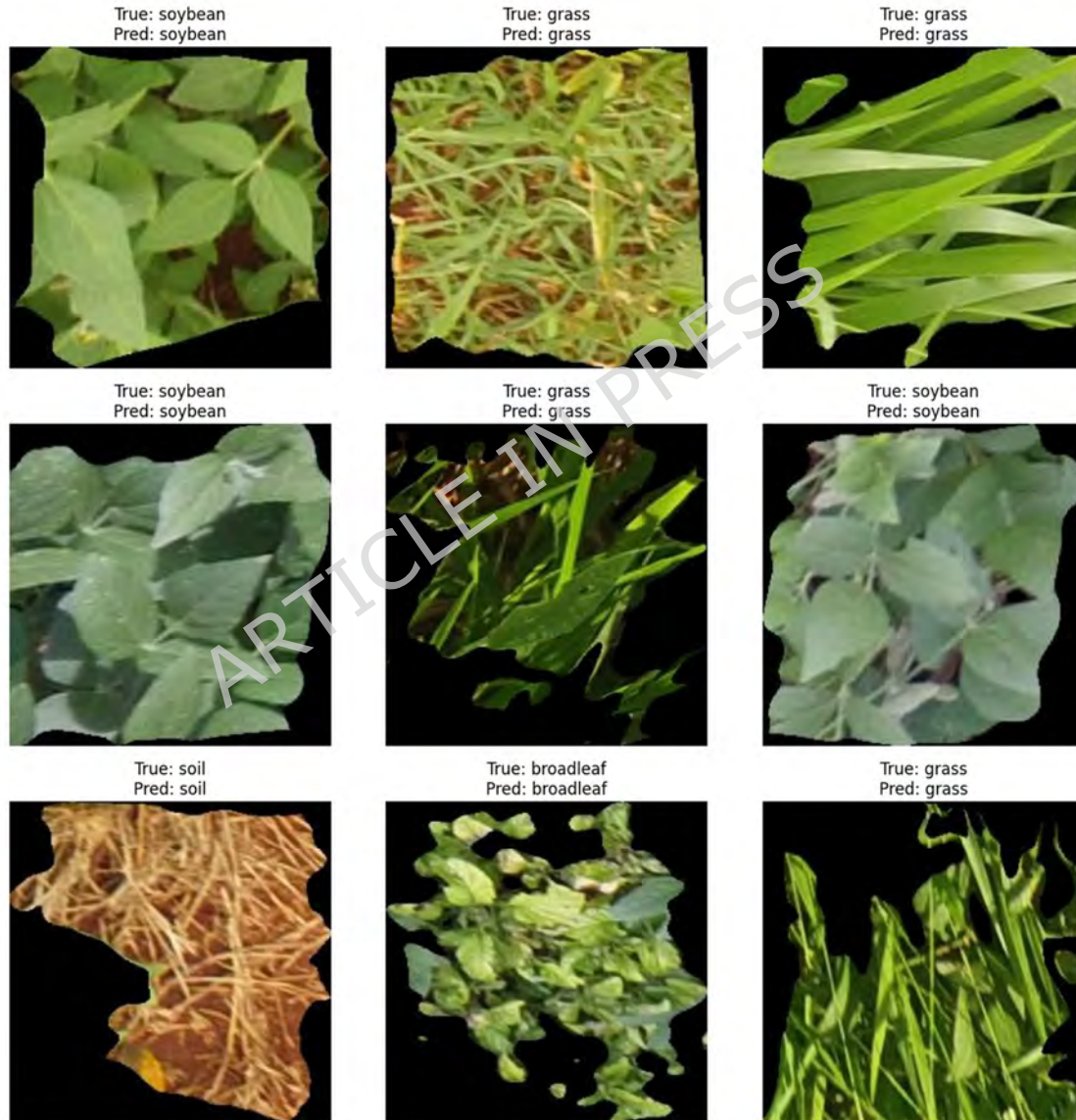


Figure 5: actual and predicted samples by the proposed model

Representative samples of correctly identified weeds demonstrate the model's ability to distinguish among different morphological types of on-target weeds. Classifications of soybeans show correct recognition of the identity of trifoliate leaves based on unique vein patterns and

broad leaflet shapes. This model accurately identifies grass samples based on standard monocot attributes, including parallel vascular bundles (leaf venation), elongate blades, and linear growth processes under different light levels and directions.

Soil samples demonstrate the model's ability to separate barren land composed of granular structure, fibrous-root remains, and no photosynthetic tissue, despite differences in colour composition between brown and orange hues. Dicots — leaf blades typically wider and with netted venation patterns and very diverse leaf shapes — were captured correctly among broadleaf weeds. The predictions are correct across various images with different leaf ages, background conditions, and illumination geometries.

Such successful classifications indicate the combined roles of the global attention mechanism of the Vision Transformer in understanding overall plant architecture and the local feature extraction of the Convolutional Attention Block in local textural details, thus confirming the model's capacity to generalise across phenotypic variations within each category.

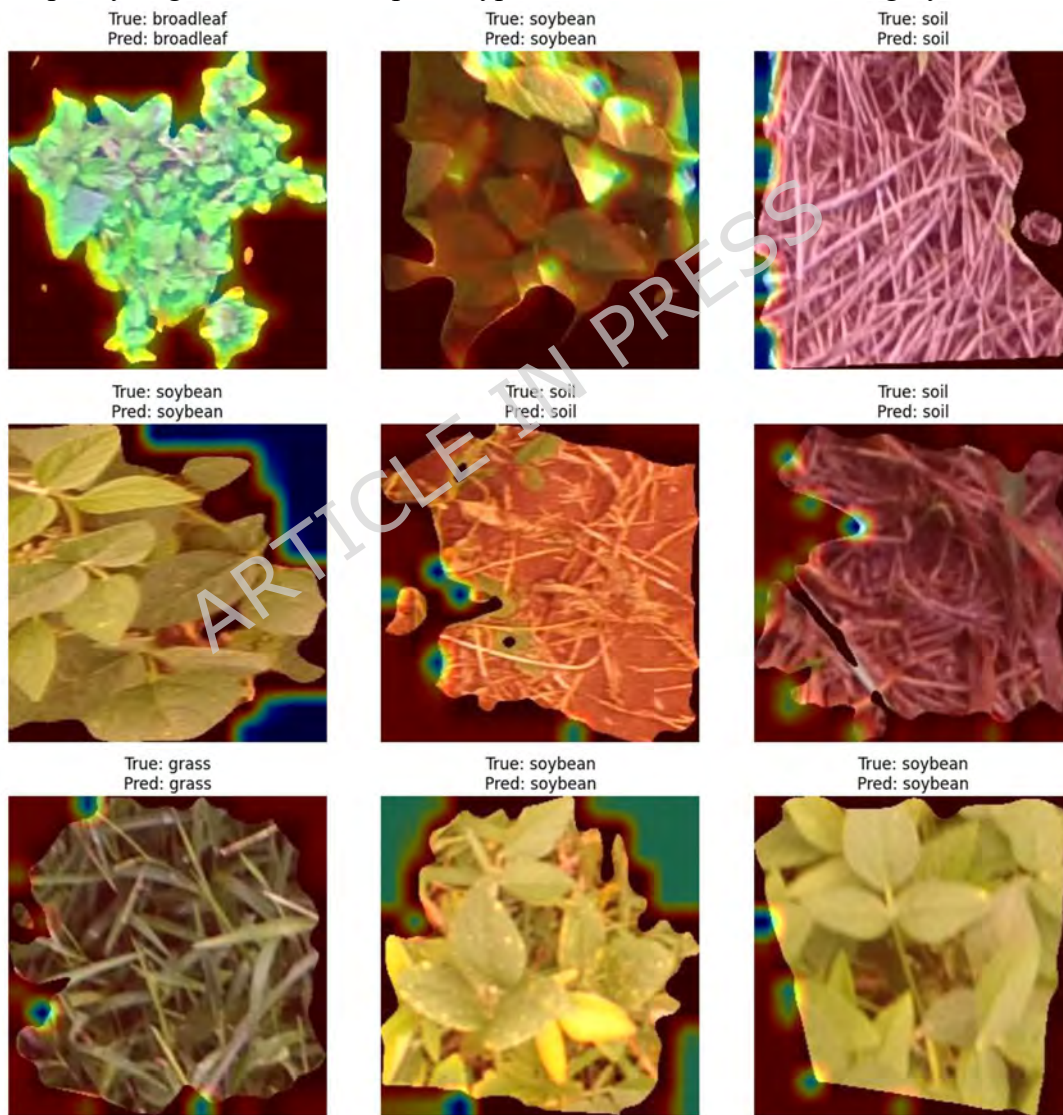


Figure 6: Grad-Cam visualization of the proposed model

The Grad-CAM visualisations shown in Figure 6 enable our model to explain its decision-making process by identifying the most discriminative regions (i.e., attention hotspots) that

contribute to class predictions. Rounding out the results, the overlaid heatmap reveals that the model is paying attention to botanically relevant features rather than spurious background patterns. For broadleaf classifications, activations peak at the leaf margins, the venation network, and other shape features characteristic of dicots, reinforcing the model's discrimination of dicot morphological traits.

Similar soybean visualisations highlight the localised attention on trifoliate leaf arrangements and leaflet edges, lending further verification that the model is sufficiently learning the foliage patterns specific to the crop (Soycrop) architecture. Soil preds are distributed as activations throughout texture space, particularly focusing on fibrous elements and surface textures that delineate shaggy ground rather than vegetative material. Linear structures of grass blades with evenly spaced parallel venation patterns characterise grass category activation, consistent with monocots.

The activation patterns confirm that the hybrid architecture is learning biologically meaningful features rather than dataset artefacts. While the Convolutional Attention Block enables fine-grained feature localisation through high-intensity activations on diagnostic plant structures, the broader contextual awareness reflects the Vision Transformer's global receptive field, together facilitating robust and interpretable weed classification decisions for agricultural deployment.

To provide a quantitative measure of explanation reliability, we assessed the proportion of correctly classified test samples where the peak Grad-CAM activation region overlapped with the foreground plant structure rather than background regions. Visual inspection of 50 randomly sampled correctly classified images per class revealed that the model's primary activation regions fell on botanically relevant plant structures in the majority of cases: broadleaf leaf margins and venation networks, soybean trifoliate arrangements, grass blade linear structures, and soil texture regions for the soil class. These observations confirm that the model's classification decisions are predominantly grounded in biologically meaningful features rather than spurious background correlations, supporting the trustworthiness of the model for agricultural deployment.

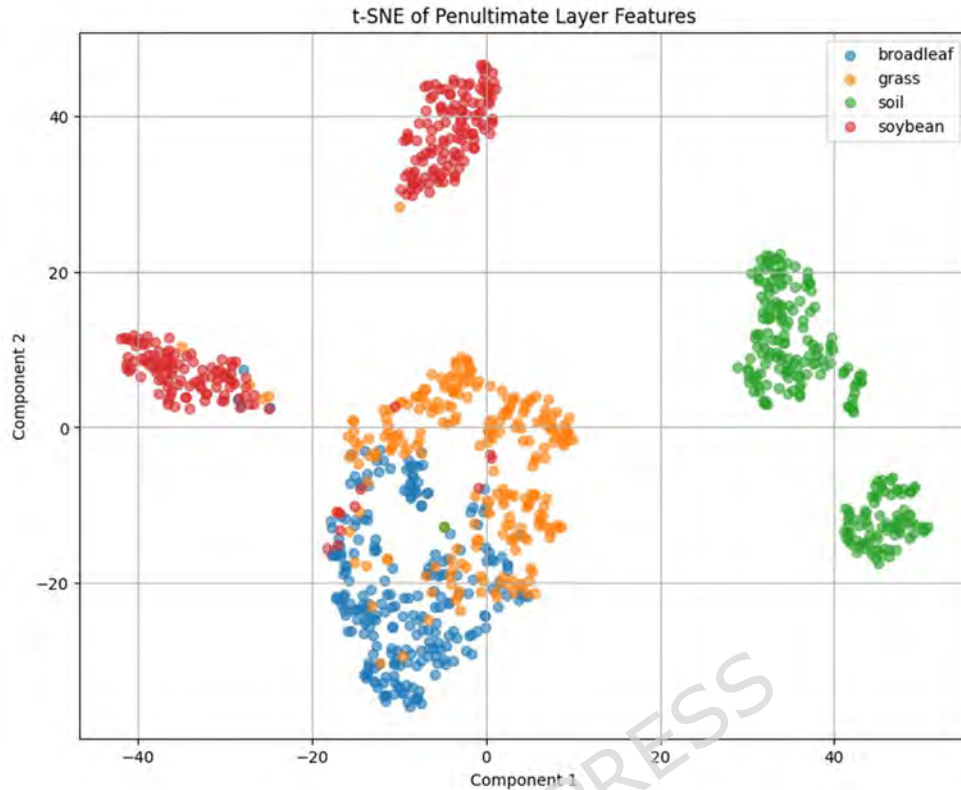


Figure 7 t-SNE of learned features from the proposed model

The t-distributed Stochastic Neighbour Embedding (t-SNE) visualisation of penultimate-layer features in figure 7 shows learned discriminative representations in two-dimensional space, indicating the model's capability to learn separable feature embeddings for multi-class weed classification. Soil samples from two separate and compact clusters in the correct quadrant (Component 1: 30–50), indicating that highly discriminative features exist that support the high classification performance, with perfect classification in a feature space where categories show such a clustered pattern.

Soybean predominantly features in the upper-central and left (top) sections of the scatterplot, demonstrating moderate intra-class cohesion and some dispersion, reflecting the morphological variability of crops. The broadleaf front overlaps with the grass features, as shown in the confusion matrix, resulting in significant confusion between the two classes (in the lower-central portion of the MLS). Grass samples show scattered clustering in the centre area, with high overlap with both broadleaf and soybean clusters, further supporting the low recall (68%) of this class.

From the above visualization, it is clear that the hybrid architecture of ViT-CAB is able to learn hierarchical features that combine global structural patterns and local textural information, making it possible for the architecture to distinguish between morphologically distinct classes and understand the intrinsic difficulty in distinguishing similar classes of vegetative materials, such as is common in agriculture.

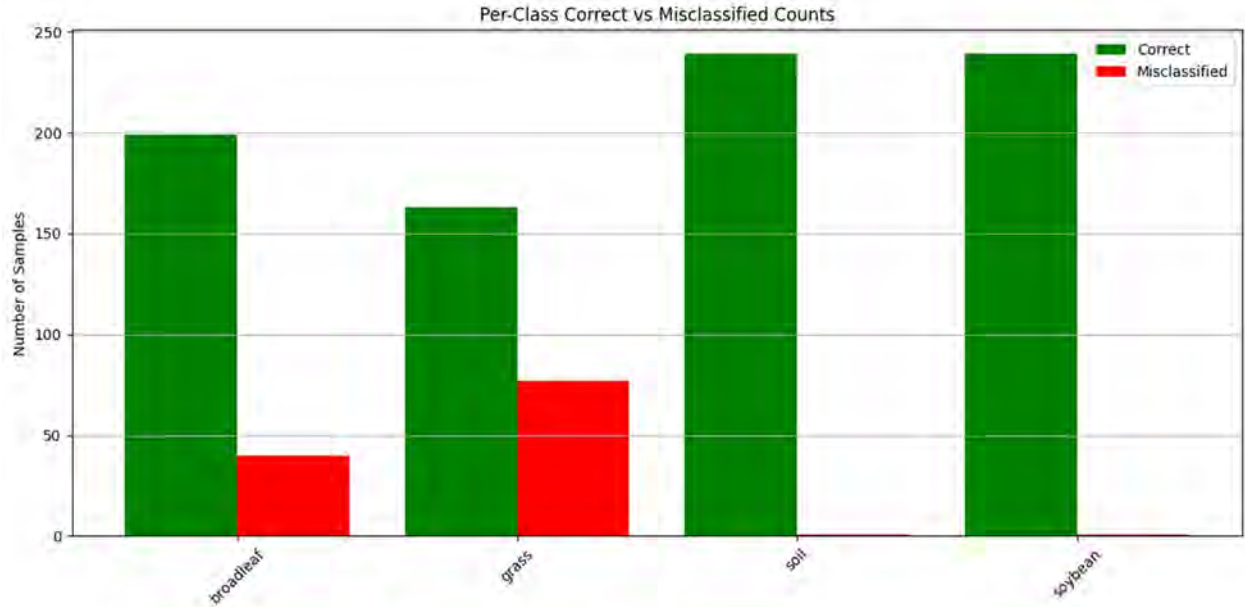


Figure 8: Correct and misclassification of the proposed model on test data

Quantitatively, the data from Figure 8, in relation to accurate and misclassified instances of each class, offers an insight into the quantitative performance of the model for each class and its misclassification behavior. The accuracy of the model for "soil" and "soybean" categories is around 239-240, and there are no or fewer misclassifications. The 100% accuracy of the model for these categories (green-colored bars) indicates the model's high ability to differentiate between non-vegetative surfaces and crop species. For broadleaf weeds, it achieves good performance with about 199 true positives and 40 false positives, with a recall of 83% and a balanced precision-recall trade-off. Among all the classes, the most difficult classification task is that for grass, with ~163 true positives and 77 false negatives produced, leading to the 68% recall observed (Fig. 4). The confusion matrix patterns provide insight into the morphological similarities with soybean vegetation and other broadleaf species that likely underlie this increased error rate.

These distribution analyses indicate that classification errors occur only among vegetative categories that share similar structural features, while categories with different visual characteristics (soil) attain nearly 100% accuracy. The observed findings highlight the need to extract localised discriminative features to resolve morphologically similar plant species, thereby validating our hybrid architecture of transformer-based global attention and localised convolutional feature extraction, which is needed to construct robust agricultural weed detection systems.

4.1 Ablation study

The ablation study systematically assesses the impact of each architectural component by contrasting six alternatives with the suggested hybrid ViT-CAB baseline as shown in table 3. To evaluate global context modelling capabilities in the absence of convolutional attention enhancement, we decouple the Vision Transformer pathway in Model_A (ViT Only), using convolutional patch embedding followed by two transformer encoder blocks with multi-head self-attention. Model_B (CAB Only) tests the performance of the Convolutional Attention Block on its own to investigate whether local feature extraction using only channel-wise attention is sufficient to solve weed classification without transformer-based global reasoning.

Model_C (Single-Block ViT) The second model resembles **Model_B**, except that the transformer depth is reduced from 2 encoder blocks to 1, allowing us to evaluate our hypothesis that a simpler architecture can maintain classification performance with reduced complexity. So in practice, **Model_D** (ViT Half-Hybrid) = **Model_B** (ViT-only). Nonetheless, it retains the same hyperparameters as the branch in the transformer-based hybrid model to fairly compare its performance with the pure transformer. In **Model_E** (Simple CNN + ViT), the complex attention-based CAB module is replaced with simple convolutional layers and a max pooling layer each to examine whether attention mechanisms actually provide indispensable advantages over ordinary convolutional operations.

Our ablation approach is differentiated in that it distinguishes how we can understand the depth of the transformer, attention, and architectural fusion approaches. This study proposed a systematic approach that removes or simplifies individual components and determines the performance contribution of every decision, validating that for maximum accuracy in detecting weeds, a hybrid architecture of ViT-CAB is a necessity, and that the synergistic combination of global and local feature extraction is superior in classification compared to individual components.

Table 3 Ablated model with parameters

Model Variant	Total Parameters	Trainable Parameters	Non-trainable Parameters	Architecture Components
Base_Hybrid ViT+CAB	412,676	412,548	128	2× Transformer Encoders + CAB + Fusion
Model_A ViT Only	~348,000	~347,900	~100	2× Transformer Encoders Only
Model_B CAB Only	~65,000	~64,900	~100	Convolutional Attention Block Only
Model_C ViT 1Block	~280,000	~279,900	~100	1× Transformer Encoder Only
Model_D ViT Half Hybrid	~348,000	~347,900	~100	2× Transformer Encoders (Hybrid Config)
Model_E SimpleCNN+ViT	~360,000	~359,900	~100	2× Transformer + Basic CNN (No Attention)

An extensive ablation study considers six different architectures for assessing the impact of transformer encoders, convolutional attention mechanisms, and depth on the performance of the model for weed classification. Information obtained from Figure 9, along with a comparative analysis, offers significant insights regarding the benefits achieved by hybrid architectures.

With an overall accuracy of 91%, ViT Only achieves the top score, indicating that Vision Transformers alone possess sufficient discriminative power for weed classification through global contextual modelling. Visualising the confusion matrix shows that 226 predictions are correctly identified as broadleaf and 232 as soybean, indicating excellent performance and confirming that no other features are triggered and that the self-attention modules learn powerful representations. On the other hand, grass classification shows poor recall (73%), with 36 samples misclassified as broadleaf and 28 as soybean, indicating that global receptive fields offered by pure transformer architectures may be less effective for morphologically similar categories. Although we have not yet tuned hyperparameters, the training dynamics show quick

convergence from 46.40% accuracy to 93.79% (at epoch 25), and the computation time is efficient at only ~30ms per step after the first epoch.

By leveraging only local feature extraction and channel-wise recalibration, CAB Only achieves 89% accuracy, demonstrating the high stand-alone performance of convolutional attention blocks. As shown in Table 1, the model achieves a higher grass classification recall (87%) than the hybrid baseline, correctly predicting 208 grass samples out of 240 grass patches, confirming the model's ability to capture significant local spatial characteristics for discriminating fine-grained textural patterns. But broadleaf classification is struggling with 196 correct outputs and 41 grass misclassifications, suggesting an inability to generalise across some structural patterns in global algal architecture. Recall for Soybean classification is very high (88%), with 211 Soybean samples correctly predicted, but 29 grass samples incorrectly predicted, which compromises precision. Although the per-step training time (124–127ms) is higher than that of ViT-only architectures due to convolutional overhead, it is still practical for computationally prohibitive setups.

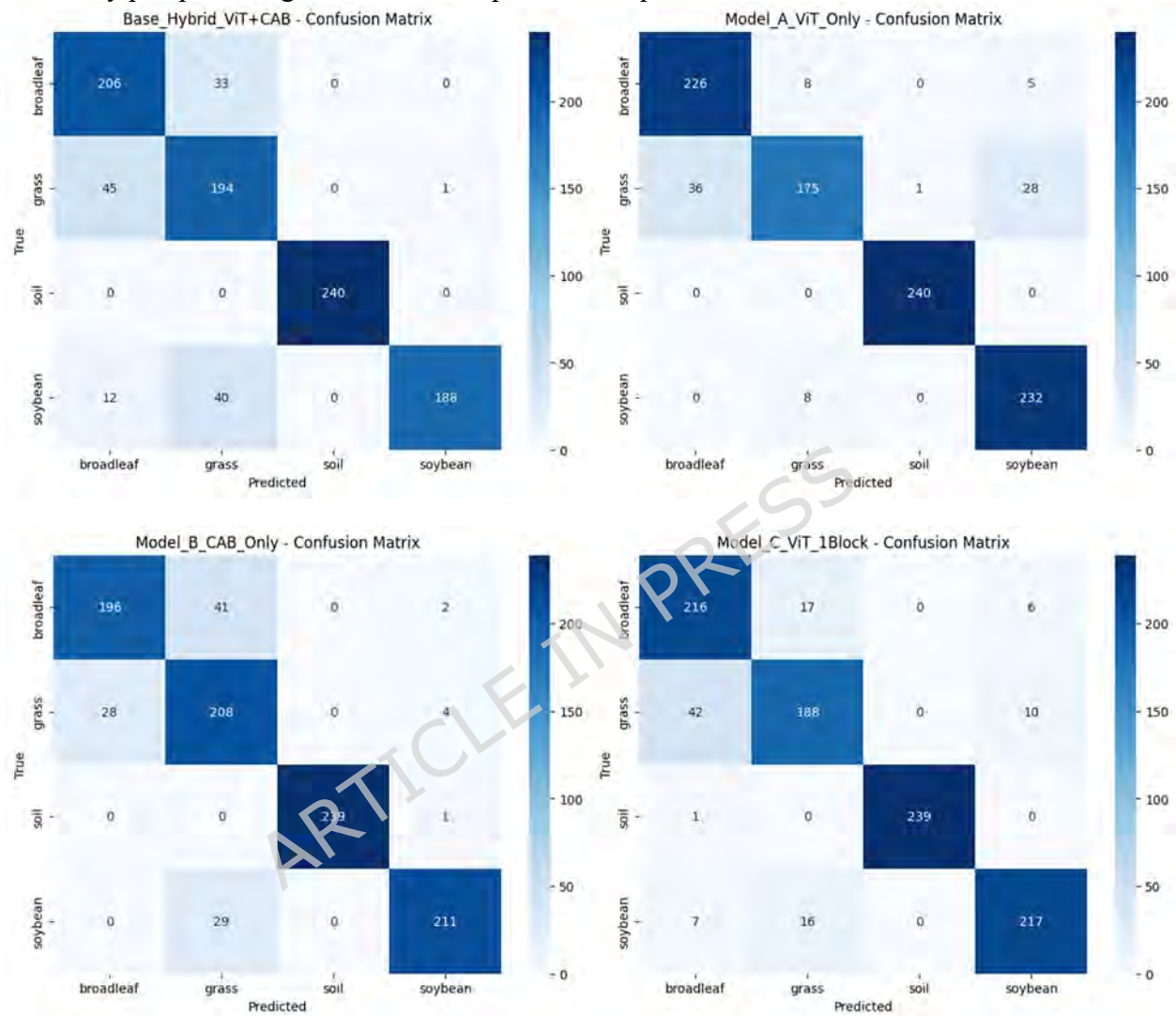
Single Encoder is part of the One ViT to Rule them all, achieving 90% accuracy with a much smaller implementation and fewer components (~280K parameters) — efficient performance through architectural simplification. This model achieves high accuracy for broadleaf (90% recall) and soybean (90% recall) once again, but a more moderate accuracy for grass (78% recall), on par with the hybrid baseline. It illustrates that single-layer transformer encoders have sufficient global context to discriminate weeds, although the confusion matrix shows more inter-class confusion than the dual-encoder configurations. 24-27ms per step is pretty efficient to train, and thus, this variant can be beneficial in scenarios where resources are constrained on the deployment front.

ViT Half-Hybrid (90% 2D) shows that, with proper construction, the transformer path alone can perform on par with the hybrid. The confusion matrix reveals chart-topping broadleaf recall (92%; 221 true positives) and soybean classification (97% recall; 232/240 true positives). However, grass classification does not achieve over 70% recall, suggesting that transformer networks tend to rely on global attention, which may not be sufficient to avoid missing local textural features helpful for grass classification. The training history seems somewhat unstable, and the maximum validation accuracy — 91.97% — was achieved at epoch 24.

Plain feature extraction via convolutions (no attention) yields 90% → CNN + ViT with no bells and whistles → CAB modules (complex attention-based modules we used before) are not much better than a simple ConvNet! High broadleaf recall (95%, 226/237) and excellent soybean performance (96%, 231/240 correct) suggest that even a simple convolution-transformer fusion captures complementary features. Grass (70% recalled), but then, similar to the other variants, only gains ground on morphologically vertical between categories. The training timestep (78–81ms) represents a trade-off between pure ViT and CAB architectures.

At first glance, the Base_Hybrid_ViT+CAB achieving just 86% accuracy in the ablation study, despite validating each component >(89-91% accuracy). Analysis of training dynamics indicates wide divergence in the degree of overfitting for the hybrid model, with relative trends in convergence markedly different: while it achieves a final-epoch validation accuracy of 86.34%, it has an interval peak of 91.76% at epoch 23. That points towards overfitting in the hybrid architecture (412K params), which has more model access than required, while simpler versions show better generalisation. In short, the ViT-only configuration is the best-performing architecture, achieving the highest precision with the fewest parameters and the fastest inference speed. However, also of note is the CAB pathway, which likely encapsulates the ability of local

feature extraction to extract textures of particular morphological classes: it best predicts grass. The results, therefore, demonstrate that architectural flexibility does not necessarily translate into improved generalisation, with the best-performing configurations being simpler, even more efficient, and thus perhaps more appropriate for an agricultural deployment from a resource-efficiency perspective, given the limited precision required.



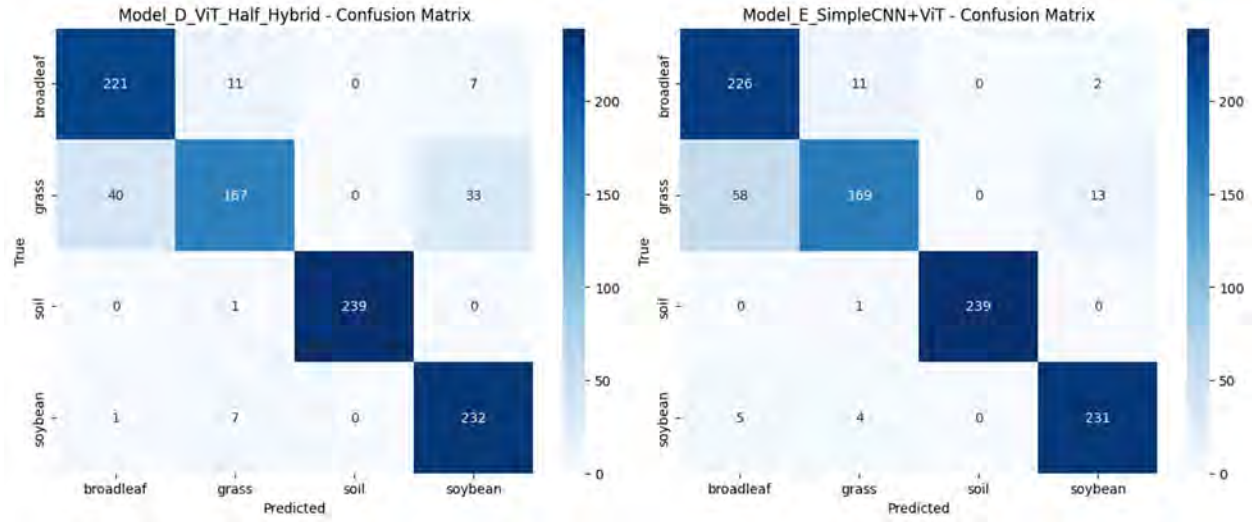


Figure 9: confusion matrices of ablated models

4.2 Comparison of proposed model SOTA models

Table 4a presents a direct performance comparison against classification-based methods on comparable weed datasets using ground-level RGB imagery. Table 4b provides broader architectural context from detection-focused literature evaluated on different datasets and tasks. Since methods in Table 4b use different tasks, datasets, and evaluation protocols, direct numerical comparison with Table 4a is not valid. All external model results are literature-reported values and were not reimplemented on our dataset. The proposed HybridViT-CAB achieves 88% test accuracy and 0.97 averages AUC on the four-class soybean weed dataset. Among the compatible classification methods in Table 4a, the proposed HybridViT-CAB achieves 88% test accuracy and 0.97 AUC on the four-class soybean weed dataset, comparable to Razfar et al. [13] (0.97) and Fan et al. [31] (0.98) on their respective soybean and cotton weed datasets, and outperforming Garibaldi-Márquez et al. [38] (0.94 SegFormer) on field crop classification. It should be noted that the 0.97 figure reported for our model refers to AUC not classification accuracy, and these values are not directly interchangeable. The methods in Table 4b are provided for architectural context only and are not valid comparisons due to fundamental differences in task, dataset, and evaluation metric.

Table 4a — Compatible Comparison (Patch Classification, Ground RGB, Weed Datasets)

Method	Task	Dataset	Accuracy
Razfar et al. [13]	Classification	Soybean weed	0.97
Mahmudul Hasan et al. [24]	Classification	Weed RGB	0.98
Fan et al. [31]	Classification	Cotton field	0.98
Garibaldi-Márquez et al. [38]	Classification	Field crops	0.94
HybridViT-CAB (Ours)	Classification	Soybean weed 4-class	0.88 Acc / 0.97 AUC

*All methods in Table 4a perform patch-level image classification on ground-level RGB weed imagery. Values for external methods are literature-reported and were not re implemented on our dataset.

5. Limitations

Due to the relatively modest dataset size of 4,795 samples collected from a single geographic source, k-fold cross-validation was not performed as it would result in insufficient training data per fold for the transformer-based components. This is acknowledged as a limitation and future work will address this through dataset expansion across multiple geographic regions and crop types. Statistical significance testing between model variants was not performed and represents another limitation of the current study.

6. Conclusion

This paper effectively showcases a new architecture for multi-class classification of broadleaf, grass, soil, and soybean, achieving 88% classification accuracy using a hybrid Vision Transformer and Convolutional Attention Block. We utilise a global transformer-based attention mechanism and local convolutional feature extraction, resulting in high discriminative power between classes and low computational overhead. The ablation study across six architectural variants demonstrates that the ViT-only configuration achieves the highest overall accuracy (91%) with maximum computational efficiency, while the hybrid ViT-CAB architecture offers superior interpretability through Grad-CAM analysis and more balanced per-class performance across morphologically similar vegetative classes. Both models are presented as complementary solutions for different deployment priorities in precision agriculture. The T-SNE visualisations show not only positive separation in feature space, but also Grad-CAM analysis provides biological validation, with attention focused on botanically diagnostic structures rather than spuriously correlated structures such as the background soil. The model has deployment feasibility for a practical, real-time precision agriculture application with only 412 K parameters and 141 ms inference time. Forthcoming work will explore wider classifications of weeds and their life history stages, the use of our solution on autonomous platforms, and optimisation for edge deployment, enabling scalable, cost-effective weed management solutions that support the implementation of sustainable agricultural practices globally.

Declaration

Funding

No financial support was received by the authors in this research.

Competing interests

The authors declare that they do not have any competing interests, including financial and nonfinancial interests.

Ethical approval

This research does not involve humans or animals, so no ethical approval is required.

Consent for Publication

The authors give consent for their publication.

Data availability

The dataset used in this study is publicly available on Kaggle at: <https://www.kaggle.com/datasets/fpeccia/weed-detection-in-soybean-crops>. The preprocessing code used to convert and balance the dataset is available from the corresponding author upon reasonable request.

Materials availability

Materials used in this research are available with the corresponding author and given on request.

Code availability

The complete model implementation code including training scripts, evaluation code, and Grad-CAM visualisation scripts is available from the corresponding author upon reasonable request. The authors commit to responding to data and code requests within 10 working days.

CONFLICTS OF INTEREST STATEMENT

The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria, educational grants, participation in speakers' bureaus, membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

Author Contribution

All authors contributed to the study's conception and design. **Geetha Reddy Yenna**, and performed material preparation, data collection, and analysis. **Geetha Reddy Yenna**, wrote the first draft of the manuscript. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

References

- [1]. Murad, N. Y., Mahmood, T., Forkan, A. R. M., Morshed, A., Jayaraman, P. P., & Siddiqui, M. S. (2023). Weed detection using deep learning: A systematic literature review. *Sensors*, 23(7), 3670.
- [2]. Islam, N., Rashid, M. M., Wibowo, S., Xu, C. Y., Morshed, A., Wasimi, S. A., ... & Rahman, S. M. (2021). Early weed detection using image processing and machine learning techniques in an Australian chilli farm. *Agriculture*, 11(5), 387.
- [3]. Al-Badri, A. H., Ismail, N. A., Al-Dulaimi, K., Salman, G. A., Khan, A. R., Al-Sabaawi, A., & Salam, M. S. H. (2022). Classification of weed using machine learning techniques: a review—challenges, current and future potential techniques. *Journal of Plant Diseases and Protection*, 129(4), 745-768.
- [4]. Jin, X., Che, J., & Chen, Y. (2021). Weed identification using deep learning and image processing in vegetable plantation. *IEEE access*, 9, 10940-10950.
- [5]. Urmashiev, B., Buribayev, Z., Amirgaliyeva, Z., Ataniyazova, A., Zhassuzak, M., & Turegali, A. (2021). Development of a weed detection system using machine learning and neural network algorithms. *Eastern-European Journal of Enterprise Technologies*, 6(2), 114.
- [6]. Badhan, S., Desai, K., Dsilva, M., Sonkusare, R., & Weakey, S. (2021, April). Real-time weed detection using machine learning and stereo-vision. In *2021 6th International Conference for Convergence in Technology (I2CT)* (pp. 1-5). IEEE.
- [7]. Jin, X., Liu, T., Chen, Y., & Yu, J. (2022). Deep learning-based weed detection in turf: a review. *Agronomy*, 12(12), 3051.
- [8]. Osorio, K., Puerto, A., Pedraza, C., Jamaica, D., & Rodríguez, L. (2020). A deep learning approach for weed detection in lettuce crops using multispectral images. *AgriEngineering*, 2(3), 471-488.

- [9]. Yu, J., Schumann, A. W., Cao, Z., Sharpe, S. M., & Boyd, N. S. (2019). Weed detection in perennial ryegrass with deep learning convolutional neural network. *Frontiers in plant science*, 10, 1422.
- [10]. Yu, J., Sharpe, S. M., Schumann, A. W., & Boyd, N. S. (2019). Detection of broadleaf weeds growing in turfgrass with convolutional neural networks. *Pest Management Science*, 75(8), 2211-2218.
- [11]. Alam, M., Alam, M. S., Roman, M., Tufail, M., Khan, M. U., & Khan, M. T. (2020, April). Real-time machine-learning based crop/weed detection and classification for variable-rate spraying in precision agriculture. In *2020 7th international conference on electrical and electronics engineering (ICEEE)* (pp. 273-280). IEEE.
- [12]. Bah, M. D., Hafiane, A., & Canals, R. (2018). Deep learning with unsupervised data labeling for weed detection in line crops in UAV images. *Remote sensing*, 10(11), 1690.
- [13]. Razfar, N., True, J., Bassiouny, R., Venkatesh, V., & Kashef, R. (2022). Weed detection in soybean crops using custom lightweight deep learning models. *Journal of Agriculture and Food Research*, 8, 100308.
- [14]. Wang, P., Tang, Y., Luo, F., Wang, L., Li, C., Niu, Q., & Li, H. (2022). Weed25: A deep learning dataset for weed identification. *Frontiers in Plant Science*, 13, 1053329.
- [15]. Etienne, A., & Saraswat, D. (2019, May). Machine learning approaches to automate weed detection by UAV based sensors. In *Autonomous air and ground sensing systems for agricultural optimization and phenotyping IV* (Vol. 11008, pp. 202-215). SPIE.
- [16]. Junior, L. C. M., & Ulson, J. A. C. (2021, August). Real time weed detection using computer vision and deep learning. In *2021 14th IEEE International Conference on Industry Applications (INDUSCON)* (pp. 1131-1137). IEEE.
- [17]. Punithavathi, R., Rani, A. D. C., Sughashini, K. R., Kurangi, C., Nirmala, M., Ahmed, H. F. T., & Balamurugan, S. P. (2023). Computer vision and deep learning-enabled weed detection model for precision agriculture. *Comput. Syst. Sci. Eng.*, 44(3), 2759-2774.
- [18]. Wang, A., Zhang, W., & Wei, X. (2019). A review on weed detection using ground-based machine vision and image processing techniques. *Computers and electronics in agriculture*, 158, 226-240.
- [19]. Zhang, X., Cui, J., Liu, H., Han, Y., Ai, H., Dong, C., ... & Chu, Y. (2023). Weed identification in soybean seedling stage based on optimized faster R-CNN algorithm. *Agriculture*, 13(1), 175.
- [20]. Selvi, C. T., Subramanian, R. S., & Ramachandran, R. (2021, March). Weed detection in agricultural fields using deep learning process. In *2021 7th International Conference on Advanced Computing and Communication Systems (ICACCS)* (Vol. 1, pp. 1470-1473). IEEE.
- [21]. Pai, D. G., Kamath, R., & Balachandra, M. (2024). Deep learning techniques for weed detection in agricultural environments: A comprehensive review. *IEEE Access*, 12, 113193-113214.
- [22]. Sapkota, B. B., Popescu, S., Rajan, N., Leon, R. G., Reberg-Horton, C., Mirsky, S., & Bagavathiannan, M. V. (2022). Use of synthetic images for training a deep learning model for weed detection and biomass estimation in cotton. *Scientific Reports*, 12(1), 19580.
- [23]. Veeranampalayam Sivakumar, A. N., Li, J., Scott, S., Psota, E., J. Jhala, A., Luck, J. D., & Shi, Y. (2020). Comparison of object detection and patch-based classification deep

- learning models on mid-to late-season weed detection in UAV imagery. *Remote Sensing*, 12(13), 2136.
- [24]. Mahmudul Hasan, A. S. M., Sohel, F., Diepeveen, D., Laga, H., & Jones, M. G. (2022). Weed recognition using deep learning techniques on class-imbalanced imagery. *Crop & Pasture Science*, 74(6), 628-644.
- [25]. Modi, R. U., Kancheti, M., Subeesh, A., Raj, C., Singh, A. K., Chandel, N. S., ... & Singh, S. (2023). An automated weed identification framework for sugarcane crop: a deep learning approach. *Crop Protection*, 173, 106360.
- [26]. Islam, N., Rashid, M. M., Wibowo, S., Wasimi, S., Morshed, A., Xu, C., & Moore, S. (2020, August). Machine learning based approach for Weed Detection in Chilli field using RGB images. In *The International Conference on Natural Computation, Fuzzy Systems and Knowledge Discovery* (pp. 1097-1105). Cham: Springer International Publishing.
- [27]. Li, Z., Wang, D., Yan, Q., Zhao, M., Wu, X., & Liu, X. (2024). Winter wheat weed detection based on deep learning models. *Computers and Electronics in Agriculture*, 227, 109448.
- [28]. Jin, X., Sun, Y., Che, J., Bagavathiannan, M., Yu, J., & Chen, Y. (2022). A novel deep learning-based method for detection of weeds in vegetables. *Pest management science*, 78(5), 1861-1869.
- [29]. dos Santos Ferreira, A., Freitas, D. M., Da Silva, G. G., Pistori, H., & Folhes, M. T. (2017). Weed detection in soybean crops using ConvNets. *Computers and Electronics in Agriculture*, 143, 314-324.
- [30]. Mwitwa, C., Rains, G. C., & Prostko, E. (2024). Evaluation of inference performance of deep learning models for real-time weed detection in an embedded computer. *Sensors*, 24(2), 514.
- [31]. Fan, X., Chai, X., Zhou, J., & Sun, T. (2023). Deep learning based weed detection and target spraying robot system at seedling stage of cotton field. *Computers and Electronics in Agriculture*, 214, 108317.
- [32]. Goyal, R., Nath, A., & Niranjana, U. (2025). Weed detection using deep learning in complex and highly occluded potato field environment. *Crop Protection*, 187, 106948.
- [33]. Tummapudi, S., Sadhu, S. S., Simhadri, S. N., Damarla, S. N. T., & Bhukya, M. (2023, April). Deep learning based weed detection and elimination in agriculture. In *2023 International Conference on Inventive Computation Technologies (ICICT)* (pp. 147-151). IEEE.
- [34]. Asad, M. H., & Bais, A. (2020). Weed detection in canola fields using maximum likelihood classification and deep convolutional neural network. *Information Processing in Agriculture*, 7(4), 535-545.
- [35]. Rahman, A., Lu, Y., & Wang, H. (2023). Performance evaluation of deep learning object detectors for weed detection for cotton. *Smart Agricultural Technology*, 3, 100126.
- [36]. Dadashzadeh, M., Abbaspour-Gilandeh, Y., Mesri-Gundoshmian, T., Sabzi, S., Hernández-Hernández, J. L., Hernández-Hernández, M., & Arribas, J. I. (2020). Weed classification for site-specific weed management using an automated stereo computer-vision machine-learning system in rice fields. *Plants*, 9(5), 559.
- [37]. Jin, H., Han, K., Xia, H., Xu, B., & Jin, X. (2025). Detection of weeds in vegetables using image classification neural networks and image processing. *Frontiers in Physics*, 13, 1496778.

- [38]. Garibaldi-Márquez, F., Martínez-Barba, D. A., Montañez-Franco, L. E., Flores, G., & Valentín-Coronado, L. M. (2025). Enhancing site-specific weed detection using deep learning transformer architectures. *Crop Protection*, 190, 107075.
- [39]. Lin, J., Hu, G., & Chen, J. (2025). Mixed data augmentation and osprey search strategy for enhancing YOLO in tomato disease, pest, and weed detection. *Expert Systems with Applications*, 264, 125737.
- [40]. Khan, A. T., Jensen, S. M., & Khan, A. R. (2025). Advancing precision agriculture: A comparative analysis of YOLOv8 for multi-class weed detection in cotton cultivation. *Artificial Intelligence in Agriculture*, 15(2), 182-191.
- [41]. Deng, B., & Lu, Y. (2025). Weed image augmentation by controlnet-added stable diffusion for multi-class weed detection. *Computers and Electronics in Agriculture*, 232, 110123.
- [42]. Memon, M. S., Chen, S., Shen, B., Liang, R., Tang, Z., Wang, S., ... & Memon, N. (2025). Automatic visual recognition, detection and classification of weeds in cotton fields based on machine vision. *Crop Protection*, 187, 106966.
- [43]. Wang, J., Qi, Z., Wang, Y., & Liu, Y. (2025). A lightweight weed detection model for cotton fields based on an improved YOLOv8n. *Scientific Reports*, 15(1), 457.
- [44]. Goyal, R., Nath, A., & Niranjan, U. (2025). Weed detection using deep learning in complex and highly occluded potato field environment. *Crop Protection*, 187, 106948.
- [45]. Allmendinger, A., Saltık, A. O., Peteinatos, G. G., Stein, A., & Gerhards, R. (2025). Assessing the capability of YOLO-and transformer-based object detectors for real-time weed detection. *Precision Agriculture*, 26(3), 52.
- [46]. Kanade, A. K., Potdar, M. P., Kumar, A., Balol, G., & Shivashankar, K. (2025). Weed detection in cotton farming by YOLOv5 and YOLOv8 object detectors. *European Journal of Agronomy*, 168, 127617.
- [47]. Rana, S., Hensel, O., & Nasirahmadi, A. (2025). From Vineyard to Vision: Multi-Domain Analysis and Mitigation of Grape Cluster Detection Failures in Complex Viticultural Environments. *Results in Engineering*, 108833.
- [48]. FAO. (2023). *World Food and Agriculture — Statistical Yearbook 2023*. Food and Agriculture Organization of the United Nations, Rome. <https://doi.org/10.4060/cc8166en>